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ESS-187

B. Tech. (New Scheme) Examination, 2025

(Fourth Semester)

SINGAL AND SYSTEM

EL-404

Time : 3 hours]

[Maximum Marks : 60

Note : Attempt all questions. All questions carry equal marks.

1. (a) Explain the different types of system. Explain them.
- (b) Explain about to Deterministic and Random signals.

Or

- (a) Explain about linear and non-linear system.
 - (b) What is convolution integral and convolution sum ?
2. (a) Explain analysis of continuous and discrete time signals.
- (b) Describe Parseval's theorem.

Or

- (a) Explain about to continuous time Fourier transform.
 - (b) State and prove sampling theorem.
3. (a) What do you mean by ROC ? Explain.

- (b) Explain the concept of controllability and observability.

Or

- (a) Explain the principle of Laplace transform.
- (b) Describe state space models in detail.
4. (a) What is the ROC and its properties ?
- (b) What is Z-Transform ? Explain.

Or

- (a) Discuss inverse z-Transform.
- (b) Describe about casualty and stability in continuous time LTI system using z-transform.

5. (a) Explain the different types of Random process.
- (b) Explain power spectral density in detail.

Or

- (a) Explain about cumulative distribution function and probability density function.
- (b) Explain Wiener-Khinchin-Einstein theorem in detail.

